ALGEBRA 1 · FOUNDATIONS OF ALGEBRA

Algebra Tiles Lab

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Algebra Tiles Lab** on Mr. K's Classroom: Algebra 1 → Foundations of Algebra → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Can 3x + 2 be simplified to 5x? Explain your thinking.

   No. An x-tile and a 1-tile are different sizes, so they are not like terms and can't be combined.

2. What do you think happens when you put a +x tile and a –x tile together?

   They cancel. Together they are worth 0 (a zero pair).

2 Explore use the tool and record what you see

Combine like terms

Build each expression on the mat, remove the zero pairs, then record what is left.

Expression you build	Zero pairs removed	Simplified
2x + 3x	0	5x
4x + 2 – x	1	3x + 2
3 – 5	3	–2
x^2 + 2x – 3x + 4	2	x^2 – x + 4
–2x + 5 + 2x – 1	3	4

Distributive property

Use the Distributive property tab. Count every tile in the rectangle.

Expression	x-tiles	1-tiles (sign)	Simplified
2(x + 3)	2	6 positive	2x + 6
3(x – 2)	3	6 negative	3x – 6
4(x + 1)	4	4 positive	4x + 4
5(x – 3)	5	15 negative	5x – 15

3 Write the rule fill in each blank from what you observed

1. Like terms have the same **variable** raised to the same **exponent**.
2. A positive tile and its matching negative tile make a **zero pair**, which is worth **0**.
3. To combine like terms, add or subtract their **coefficients** and keep the variable part the same.

4. $a(x + c) =$ **ax** + **ac**, because there are a rows and each row holds one x and c ones.

4 Check no tool: show your work

1. Simplify $7x - 2 + 3x + 9$.

   $10x + 7$

2. Simplify $-4y + 6 - y - 6$.

   $-5y$

3. Simplify $3(x - 4)$.

   $3x - 12$

4. Simplify $2x^2 + 5x - x^2 - 8x$.

   $x^2 - 3x$

5 Reflect

A classmate says $2x + 3x^2 = 5x^3$. Use tiles to explain the mistake.

An x-tile and an x^2-tile are different shapes, so they are not like terms. The expression is already simplified: $3x^2 + 2x$.